

Data Science Capstone - Weekly Report 1

Project: MoodBand: An Inclusive Wearable System for Real-Time Mood Detection and Adaptive Sensory Feedback

Supervisor: Prof. Rached Zantout

Student: Ritik Pravin Mota

Meeting Info:

- Date: 21 September, 2026, Monday
- Attendees: Prof. Rached Zantout, Ritik Mota
- Mode: In person

Last Week Progress / Topics Discussed:

- As this was the first meeting for the project, there is no progress to be reported.
- Prof. Zantout walked me through the problem statement, objectives and the scope of the project.
- There has been some primitive literature review performed on the project by AI systems, which will need review and further literature review to be performed by me.
- The original scope of the project included building a prototype wearable device that would detect physiological markers to detect the mood of the user and also give appropriate feedback using vibration motors and sound.
- Since there is only one student working on the project, the scope of the project has been revised to work with existing wearables and a mobile application instead.
- The project entails building a lightweight classifier to detect moods and the mobile application to complete the feedback cycle in an attempt to improve their mood.
- By the end of the semester, we are aiming at having a working prototype application that would run on the users mobile phone for a working demo.
- Since the system is designed with accessibility in mind, there would be multiple feedback systems (auditory, haptic, visual) for impaired users.
- The professor has shared the proposal document with me which contains a detailed overview of the project.

Plan for next week:

- Perform literature review to look for existing research in the field and research gaps that can be filled.
- Find out what physiological markers can be used to detect a users mood.
- Finalise on an existing wearable that can be used for the project.

To-Do List:

- Edit the proposal document to revise the scope of the project.
- Perform literature review.